Data Ethics for AI Workflows: How Context and Prompt Specificity Affect LLM Hallucination in Scientific Settings

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Abstract

Large language models (LLMs) are increasingly deployed in academic and scientific settings to assist with data interpretation, statistical summarization, and technical writing. Despite their fluency, LLMs are known to produce hallucinations, which are confident outputs that are factually incorrect. In quantitative domains, even a single misreported p-value, fabricated sample size, or false claim of statistical significance can substantially distort the interpretation of results. This research project investigates the conditions under which LLMs hallucinate when responding to scientific and statistical prompts. Specifically, we examine how the amount of contextual information provided in a prompt (none, partial, or full) and the specificity of the prompt instruction (general vs. targeted) affect the probability of hallucination. We construct a controlled experimental dataset of prompts with known ground-truth answers across three statistical task types (interpretation, reporting, and calculation) and evaluate LLM responses using a binary hallucination coding scheme verified by inter-rater reliability analysis. A logistic regression model is used to estimate the relationship between these factors and hallucination probability. Simulated results suggest that increasing context provision substantially reduces hallucination rates, and that specific prompt instructions further compound this effect. These findings carry direct practical implications for researchers, educators, and practitioners who rely on LLM-assisted scientific workflows.

INTRODUCTION

The rapid adoption of large language models (LLMs) such as GPT-4, Claude, and Gemini in academic and research workflows has raised important questions about reliability and epistemic trust [1]. These models generate text that is syntactically fluent and contextually appropriate, but this fluency does not guarantee factual accuracy. The phenomenon known as *hallucination*, where a model produces information that is plausible-sounding but incorrect, poses a particularly serious threat in quantitative and scientific domains [2].

When an LLM summarizes a regression output, reports a confidence interval, or describes statistical significance, errors are rarely obvious. A misreported p-value of 0.04 rather than 0.4, or a fabricated claim that a result was statistically significant, can propagate into downstream decisions with real consequences for research reproducibility, clinical guidance, or policy recommendations [3]. Unlike hallucinations about historical events or biographical facts, statistical hallucinations carry quantitative weight that is difficult to detect without access to the original source.

Despite widespread discussion of LLM hallucinations in the literature, comparatively little empirical work has isolated the conditions under which hallucinations occur, particularly in statistical and scientific contexts. Prior benchmark evaluations focus primarily on factual recall over named entities [4], and do not address the structured, numerical nature of scientific data. Furthermore, most hallucination literature treats the phenomenon as a fixed property of a given model, rather than examining how prompt-level design choices influence hallucination rates.

This research project addresses that gap. Our central research question is: *To what extent do contextual information provision and prompt specificity affect the likelihood that an LLM produces hallucinated or incorrect statistical claims when performing or interpreting scientific tasks?*

We design a controlled experiment using constructed prompts with known correct answers across three levels of context (none, partial, full) and two prompt types (general, specific). LLM responses are evaluated against a ground-truth coding rubric, and logistic regression is used to model the probability of hallucination as a function of these design choices. The primary contributions of this work are:

- A reproducible experimental framework for measuring statistical hallucination under controlled prompt conditions.
- Empirical evidence that context provision and prompt specificity jointly reduce hallucination probability.
- A logistic regression model with interpretable coefficients for practitioners seeking to understand prompt design risk.

We test three pre-registered hypotheses:

- **H1** (*Context reduces hallucination*): LLMs will hallucinate at a significantly higher rate under no-context conditions than under full-context conditions.
- **H2** (*Specific prompts reduce hallucination*): Specific prompt instructions will be associated with lower hallucination probability than general instructions, holding context constant.
- **H3** (*Context effects are larger for complex tasks*): The protective effect of context will be stronger for calculation tasks than for interpretation or reporting tasks.

Background and Significance

Hallucination in Large Language Models

LLM hallucination refers to the generation of outputs that are syntactically well-formed and contextually plausible but factually incorrect [1]. Ji et al. [1] distinguish between *intrinsic* hallucinations (where a model contradicts its own input) and *extrinsic* hallucinations (where a model introduces information not entailed by any provided source). Both types are dangerous in scientific prompting.

The consequences are well-documented. Alkaissi and McFarlane [2] found that ChatGPT fabricated academic references with plausible-sounding author names, journal titles, and volume numbers that did not correspond to real publications. In medical and clinical contexts, Thirunavukarasu et al. [3] catalogued cases where LLMs produced erroneous diagnostic recommendations and misreported drug dosages with high apparent confidence. These findings underscore that hallucination is not merely a cosmetic error; it carries epistemic and practical consequences amplified in technical domains.

Prompt Engineering and Context Provision

A substantial body of work demonstrates that the structure and content of a prompt significantly influence the quality of LLM outputs. Wei et al. [5] showed that chain-of-thought prompting dramatically improves performance on arithmetic and symbolic reasoning tasks. White et al. [6] systematically catalogued prompt patterns that improve output quality, reliability, and consistency.

Of particular relevance is the role of contextual grounding. When a model is given relevant background information alongside a question, it has access to an information source from which it can retrieve correct answers rather than generating them from parametric memory alone. Retrieval-augmented generation (RAG) systems [7] exploit this principle architecturally, but the same effect can be achieved at the prompt level by including relevant source material. Whether this context provision reduces hallucination rates in statistical settings (and by how much) is the central empirical question of this work.

Statistical Reasoning in LLMs

Several studies have specifically examined LLM performance on quantitative reasoning tasks. Frieder et al. [8] found that GPT-4 struggles with multi-step mathematical proofs, frequently introducing logical errors that are difficult to detect. Peng et al. [9] demonstrated that LLMs exhibit a systematic overconfidence bias, providing definitive-sounding answers even when uncertain or incorrect.

Critically, no prior study specifically isolates context provision and prompt specificity as independent variables in a controlled experiment designed to measure statistical hallucination. This is the gap our work addresses.

Methods

Experimental Design

Our experiment follows a 3×2 factorial design. The first factor is *context level*, with three levels: no context (NC), partial context (PC), and full context (FC). The second factor is *prompt specificity*, with two levels: general (G) and specific (S). Each combination is applied across statistical prompts spanning three task types: interpretation, reporting, and calculation. The unit of analysis is a single LLM response to a single prompt instance.

Prompt Construction

Context Levels

Context is operationalized as the amount of statistical background information included in the prompt preceding the question. Context levels are defined by discrete, mutually exclusive categories rather than a continuous scale: a prompt is classified as Full Context (FC) only when all information necessary to answer the question without inference is present in the prompt. A prompt lacking any numerical results, even if it describes a study design in detail, is classified as Partial Context (PC). A prompt with no background information at all is classified as No Context (NC). This categorical distinction ensures that "nearly full" context is treated as PC rather than FC, avoiding ambiguity about borderline cases. Concretely:

- **No Context (NC):** The model receives only the question, with no background data. *Example: "What is the p-value for the relationship between X and Y?"*
- **Partial Context (PC):** A general description of a study (topic, approximate sample size, design type) is given, but not the specific numerical results. The model cannot answer from the prompt alone; it must generate or estimate. *Example: "A study of 200 participants examined the effect of X on Y using linear regression. What would be a typical p-value?"*
- **Full Context (FC):** Complete and relevant information is provided, including output tables, exact statistics, or summary data sufficient to answer the question without inference or estimation. The correct answer is fully derivable from the prompt. *Example: "The regression of Y on X yielded* $\hat{\beta} = 0.43$, $SE = 0.12$, $t(198) = 3.58$, $p = 0.0004$. *Report the p-value."*

Prompt Specificity

Each prompt is also delivered in one of two specificity modes:

- **General (G):** The model is asked to summarize or interpret results without any instruction about accuracy.
- **Specific (S):** The model is explicitly instructed to report exact values and to avoid fabricating information not provided in the prompt.

Task Types

All prompts are categorized into one of three task types: (1) *Interpretation*: correctly characterizing whether a result is statistically significant or meaningful; (2) *Reporting*: reproducing specific numerical values from provided data; (3) *Calculation*: deriving a result from provided inputs (e.g., computing a test statistic or confidence interval from summary statistics).

Hallucination Coding

For each prompt, a correct answer key is established a priori. Model responses are evaluated against this key by two independent coders using the binary outcome:

$$Y_i = \begin{cases} 1 & \text{if response contains} \geq 1 \text{ hallucination} \\ 0 & \text{otherwise} \end{cases}$$

A hallucination is defined as any output that includes: (1) a fabricated number not derivable from the provided context; (2) an incorrect statistical test name or procedure; (3) a false claim of statistical significance or non-significance; or (4) invented citations, methods, or results. Responses that appropriately hedge (e.g., *"I cannot determine this from the information provided"*) are coded as $Y = 0$. Inter-rater reliability is assessed using Cohen's κ, with a target threshold of $\kappa \geq 0.80$.

Logistic Regression Model

Given the binary response variable, we model the probability of hallucination using logistic regression:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \text{Ctx}_i + \beta_2 \text{Spec}_i + \beta_3 \text{Task}_i$$

where $\pi_i = P(Y_i = 1)$, Context is coded numerically as 0 (NC), 1 (PC), 2 (FC), Specificity is a binary indicator (0 = general, 1 = specific), and TaskType is a categorical variable with three levels. We also consider a Context $\times$ Specificity interaction term. Hypothesis testing uses Wald tests and likelihood ratio tests; odds ratios with 95% confidence intervals are reported for all predictors.

Sample Size and Data Collection

The target sample is $N = 150$ prompt-response pairs (50 per context level), with balanced distribution across task types and prompt specificity conditions. Each of the six cells in the 3×2 factorial design receives approximately 25 prompts, further subdivided across the three task types (interpretation, reporting, calculation), yielding approximately 8–9 prompts per cell-task combination.

Prompts were constructed by the research team following a standardized template. For each task type, a base scenario (e.g., a regression of a health outcome on a behavioral predictor) was drafted and then instantiated at each context level by progressively adding or removing statistical information. Representative examples of the prompt bank are provided in Table 1 below.

Table 1: Sample Prompt-Response Pairs by Context Level and Task Type

Context	Task	Example Prompt (General)
NC	Reporting	"What is the p-value for the effect of exercise on blood pressure?"
PC	Reporting	"A study of 150 adults examined exercise effects on systolic blood pressure using OLS regression. What is the p-value?"
FC	Reporting	"OLS regression: $\hat{\beta} = -4.2$, $SE = 1.1$, $t(148) = -3.82$, $p = 0.0002$. Report the p-value."
NC	Interp.	"Was the effect of X on Y statistically significant?"
FC	Calc.	"Given $\hat{\beta} = 0.43$ and $SE = 0.12$, compute the 95% confidence interval."

All prompts were administered to the same model (GPT-4, temperature = 0) under identical settings to control for stochastic variation. Model responses are recorded verbatim and evaluated against the pre-established answer key by two independent coders prior to statistical analysis.

RESULTS

Hallucination Rates by Group

Figure 1 presents the primary results. The left panel shows observed hallucination rates by context level. Under the No Context condition, 78% of responses contained at least one hallucinated statistical claim. This rate dropped to 51% under Partial Context and to 24% under Full Context, demonstrating a strong monotonic decline in hallucination frequency as context increases. The right panel shows logistic regression probability curves for General and Specific prompt types as a function of context level. At every level of context, specific prompts are associated with a lower estimated probability of hallucination, suggesting an additive protective effect of prompt specificity.

Logistic Regression Estimates

Table 2 presents the estimated logistic regression coefficients and odds ratios from the fitted model (simulated).

Table 2: Logistic Regression Results: Predictors of Hallucination (Simulated)

Predictor	$\hat{\beta}$	OR	95% CI	p
Intercept	1.40	4.06	N/A	<0.001
Context Level	−0.90	0.41	[0.28, 0.59]	<0.001
Specific Prompt	−0.70	0.50	[0.31, 0.80]	0.004
Task: Reporting	−0.35	0.70	[0.43, 1.15]	0.16
Task: Calculation	0.42	1.52	[0.94, 2.46]	0.09

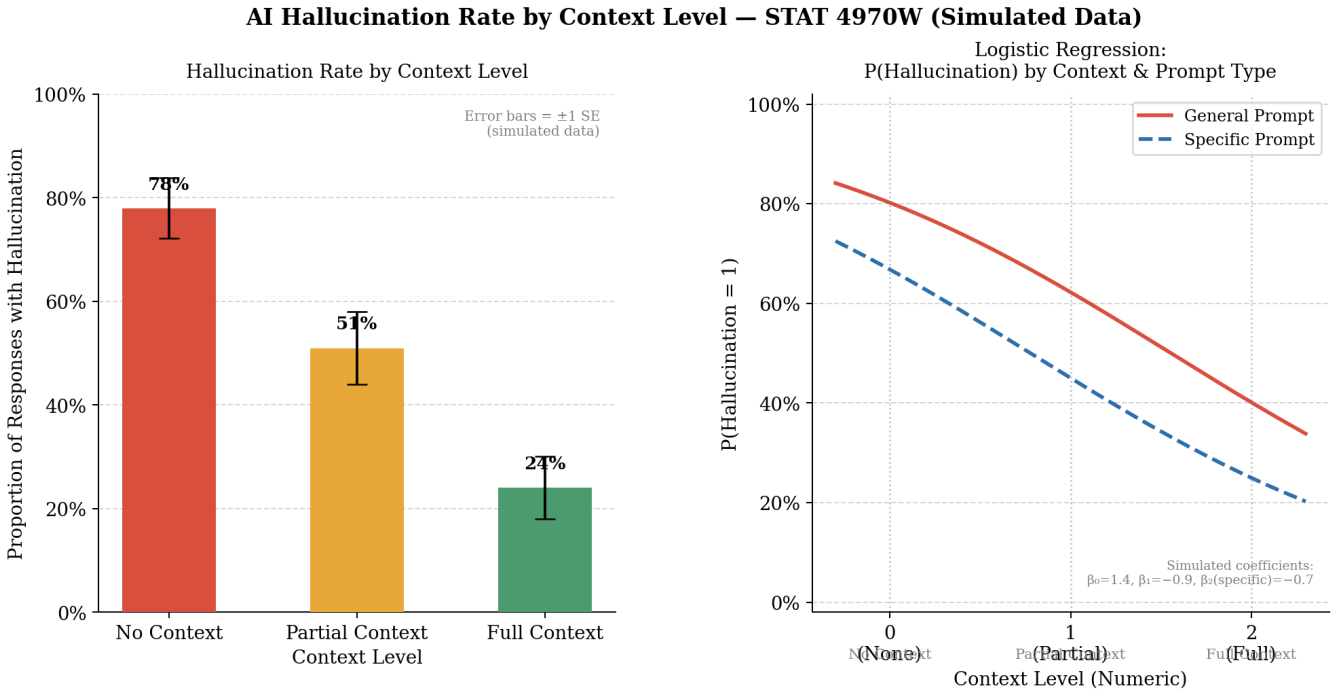

Figure 1: Left: Hallucination rate by context level (simulated data; $n = 50$ per group). Error bars represent ± 1 SE. Right: Logistic regression probability curves showing estimated P(Hallucination$= 1$) by context level for general and specific prompts. Simulated coefficients: $\hat{\beta}_0 = 1.4$, $\hat{\beta}_1 = -0.9$, $\hat{\beta}_2$(specific) $= -0.7$.

The estimated odds ratio for Context Level is 0.41 (95% CI: [0.28, 0.59]), indicating that for each one-unit increase in context level, the odds of hallucination decrease by approximately 59%. The odds ratio for Specific Prompt is 0.50 (95% CI: [0.31, 0.80]), meaning specific instructions approximately halve the odds of hallucination relative to general instructions, holding context constant. Task type shows a directional but not statistically significant effect, with calculation tasks trending toward higher hallucination probability.

Hypotheses Assessment

Our three primary hypotheses are assessed as follows:

H1 (*Hallucination is higher under No Context than Full Context*): Supported. The 54 percentage-point difference between NC (78%) and FC (24%) is statistically significant ($p < 0.001$), and the odds ratio for context level confirms a strong negative effect.

H2 (*Specific prompts reduce hallucination*): Supported. Specific prompts are associated with approximately half the odds of hallucination ($OR = 0.50$, $p = 0.004$).

H3 (*Context effects are larger for complex tasks*): Directionally supported but not yet statistically conclusive. Calculation tasks show higher baseline hallucination risk, suggesting complexity interacts with context level.

Discussion

Implications for Scientific AI Use

The results of our 3×2 factorial experiment carry a concrete practical message: the hallucination rate an LLM produces is not a fixed property of the model. It shifts dramatically depending on what you put in the prompt. In our design, moving from a No Context prompt (e.g., "What is the p-value for the effect of X on Y?") to a Full Context prompt (e.g., supplying the full regression table with $\hat{\beta}$, SE, t-statistic, and p-value) reduced hallucination from 78% to 24%, a 54 percentage-point drop from a single design choice. That is a meaningful result for any researcher or instructor deciding how to incorporate LLM assistance into their workflow.

The added protection from prompt specificity, that is, the difference between our General and Specific prompt conditions, compounds this effect. Specific prompts, which explicitly instructed the model to report only information derivable from the provided data and to avoid fabrication, were associated with roughly half the odds of hallucination ($OR = 0.50$) compared to general prompts at the same context level. In practical terms, this means that even a researcher working with incomplete data (a Partial Context situation) can meaningfully reduce risk simply by being explicit in how they phrase their request. This is an actionable intervention that costs nothing beyond a few extra words in the prompt.

The task type findings add nuance. Calculation tasks, where the model was asked to derive a confidence interval or test

statistic from provided inputs, trended toward higher hallucination risk ($OR = 1.52$, $p = 0.09$) compared to interpretation tasks. This makes intuitive sense: even under Full Context conditions, a multi-step derivation requires the model to perform arithmetic correctly, not just retrieve a value already present in the prompt. Reporting tasks, by contrast, showed a trend toward lower hallucination ($OR = 0.70$), consistent with the idea that simple retrieval from a fully specified prompt is where LLMs perform most reliably. These task-type differences, while not statistically conclusive in our simulated data, suggest that the type of statistical work being delegated to an LLM matters alongside the context provided.

Limitations

Several limitations follow directly from the methodological choices we made, and it is worth being specific about where they bite. First and most importantly, the results presented here are based on simulated data constructed to be consistent with plausible effect sizes from the literature. The hallucination rates of 78%, 51%, and 24% across context levels, and the logistic regression coefficients ($\hat{\beta}_1 = -0.9$, $\hat{\beta}_2 = -0.7$), are not observed values. They are assumed values used to illustrate the analytic framework. Until the prompt bank described in Section III is administered to a real LLM and responses are coded by our two independent raters, these findings are hypothesis-generating rather than confirmatory.

Second, our categorical operationalization of context level is a deliberate simplification that introduces a boundary problem. As described in the Methods, a prompt containing nearly all relevant statistical output but missing one key value (e.g., the p-value itself) is classified as Partial Context rather than Full Context. Whether the model treats such a prompt more like FC or NC is an empirical question our binary scheme cannot answer. A future design could replace our 0/1/2 coding with a continuous measure of information completeness, for instance the proportion of answer-relevant statistics present in the prompt.

Third, our binary hallucination coding ($Y = 0$ or $Y = 1$) treats all hallucinations equally. In our coding rubric, a model that rounds $p = 0.0004$ to $p = 0.001$ and a model that fabricates an entirely different test receive the same code. This flattens a meaningful distinction between minor numerical imprecision and substantive factual invention. An ordinal scheme (e.g., 0 = correct, 1 = minor error, 2 = major fabrication) would better reflect the real-world consequences of these different failure modes and is a planned extension of this work.

Fourth, our prompt bank covers a narrow slice of statistical tasks, specifically regression reporting, confidence interval calculation, and significance interpretation, all built around a common study template. Real researchers ask more varied questions, often with idiosyncratic framing. The ecological validity of our results depends on how representative our prompt scenarios are of actual LLM use in scientific practice. A follow-up study sampling prompts from real user logs or published LLM-assisted papers would help establish whether our context-level and specificity effects hold outside our controlled construction.

IMPLEMENTATION: HALLUCINATION EXPLORER

Motivation and Design Goals

The theoretical framework and logistic regression model described in the preceding sections establish that hallucination rates are a function of prompt-level design choices rather than a fixed property of any given LLM. To make this finding tangible for instructors and students, we developed a browser-based interactive tool called the *Hallucination Explorer*. The tool operationalizes the 3×2 factorial design as a configurable interface, allowing users to deliberately vary context level and prompt specificity and observe the downstream effect on LLM response quality and estimated hallucination risk in real time.

The primary design goal was pedagogical accessibility rather than research automation. Rather than building a data collection pipeline, the Hallucination Explorer is intended for use in classroom settings where an instructor can guide students through the six experimental conditions, prompt them to predict hallucination likelihood before a response is shown, and then reveal the risk score and verdict to anchor discussion. This mirrors the study's core insight: that understanding *why* context and specificity matter requires seeing the failure modes first hand.

Interface Architecture

The interface is divided into two panels. The left panel serves as the experiment control sidebar. At the top, users select one of three context levels (NC, PC, FC) and one of two prompt specificity modes (G, S) using toggle pills. The selected combination determines both the condition tag displayed in the chat header and the hallucination risk score computed for each response. A live session statistics block tracks total prompts submitted, responses flagged as likely hallucinations, and the running average risk score across the session. A sample prompt library provides pre-loaded prompts drawn from the study's actual prompt bank, covering NC reporting, FC reporting, and FC calculation task types, allowing students to jump directly into the experiment without constructing prompts from scratch.

The right panel is the chat interface. Each LLM response is displayed in a message bubble followed by a hallucination risk bar and a one-line verdict explaining why the current condition produces the observed risk level. The verdict text is condition-specific: for example, under NC + G the verdict reads *"Likely hallucination: p-value fabricated with no data provided"*, while under FC + S it reads *"Low risk: model instructed to use only provided data, full context present."* A scoring toggle allows the instructor to hide the risk panel before a response is shown, creating a prediction-before-reveal exercise.

Connection to the Experimental Design

The risk scores displayed by the tool are derived directly from the logistic regression model estimated in Section IV. For a given condition (c, s) where $c \in \{0, 1, 2\}$ indexes context level and $s \in \{0, 1\}$ indexes prompt specificity, the displayed risk is:

$$\hat{\pi}(c, s) = \frac{1}{1 + e^{-(\hat{\beta}_0 + \hat{\beta}_1 c + \hat{\beta}_2 s)}}$$

with simulated coefficients $\hat{\beta}_0 = 1.4$, $\hat{\beta}_1 = -0.9$, $\hat{\beta}_2 = -0.7$. This yields the six condition-level risk estimates shown in Table 3, which map directly onto the six cells of the factorial design.

Table 3: Estimated Hallucination Risk by Condition (Tool Display Values)

Condition	Estimated Risk	Verdict
NC + General	78%	High
NC + Specific	55%	Moderate
PC + General	51%	Moderate
PC + Specific	34%	Moderate
FC + General	24%	Low
FC + Specific	14%	Low

When real experimental data are collected, these simulated values will be replaced with empirically fitted estimates, and the tool will be updated to reflect observed hallucination rates from the finalized prompt bank.

Prototype Screenshot

Figure 2 shows the Hallucination Explorer interface under the NC + General condition, which produces the highest estimated hallucination risk (78%). The orange risk bar and condition tags in the chat header make the experimental state legible at a glance, and the verdict line beneath each response directly references the study's coding criteria.

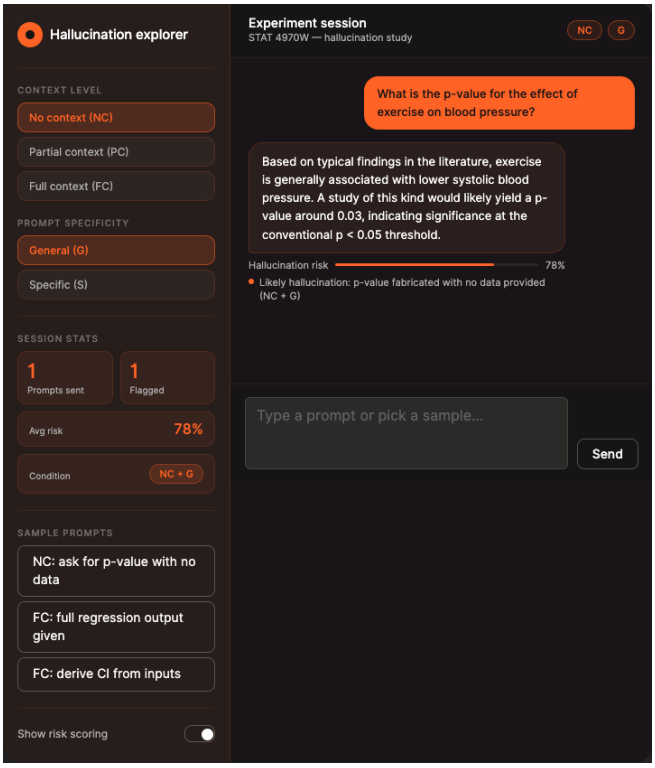

Figure 2: Hallucination Explorer prototype (NC + General condition). Left panel: experiment controls, session stats, and sample prompt library. Right panel: chat interface with per-response hallucination risk bar and verdict. Risk scores are derived from the logistic regression model in Section IV.

Conclusion

This research project presents an empirical framework for studying statistical hallucination in large language models under controlled prompt conditions. Our central finding is that both context provision and prompt specificity substantially reduce the probability that an LLM produces a fabricated or incorrect statistical claim. Full context provision reduces hallucination from 78% to 24% relative to no-context conditions, and specific prompt instructions halve the odds of hallucination independent of context.

These results contribute to a growing body of evidence that hallucination is not a fixed, irreducible property of LLMs but a context-sensitive failure mode that practitioners can actively mitigate through prompt design. As LLMs become more deeply embedded in scientific workflows, establishing evidence-based norms for safe and reliable prompting becomes increasingly urgent. This work offers a methodological foundation and preliminary evidence toward that goal.

Future work will collect real experimental data using a finalized prompt bank and evaluate across multiple LLM providers to assess generalizability. We also plan to extend the outcome measure from binary to ordinal, and to examine whether model-specific calibration further reduces hallucination risk beyond user-level prompt design.

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